

Chi-square Tests

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Hypothesis testing: null and alternative ([Note 291](#)); Joint and marginal probability ([Note 341](#)).
- **Compare with:** One-sample proportion test ([Note 570](#)).

1. Overview

Key point: A chi-square test compares the counts we observe in each category with the counts we would expect if H_0 were true; the bigger the total mismatch χ^2 , the smaller the p-value.

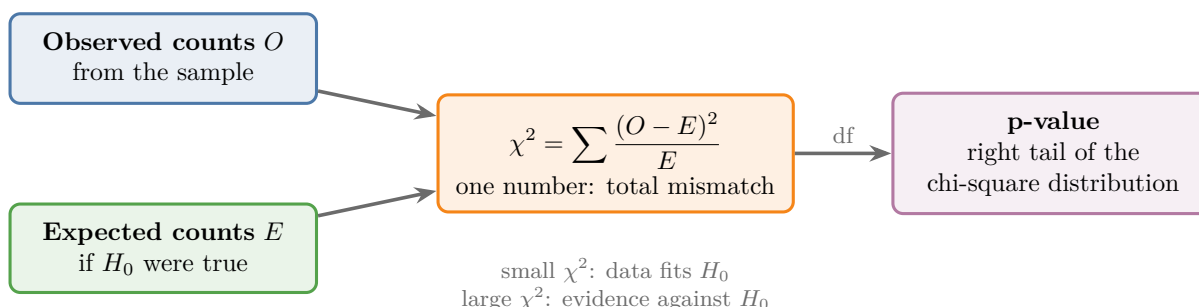


Figure 1: The chi-square test in four steps

The chi-square test was named in the [what is statistics Note](#) as the test for categorical data. In the [choosing a hypothesis test Note](#) the chi-square test is the test for two categorical **features** (a feature is a variable of the data, one column of the data table, such as gender). Figure 1 shows how every chi-square test works: observed counts and expected counts go into one number, χ^2 , and its tail area under the chi-square distribution is the p-value.

This Note covers:

- the chi-square statistic and the chi-square distribution;
- the **goodness-of-fit test**: does one categorical feature match claimed proportions?
- the **test of independence**: are two categorical features related?
- a case study on the Titanic, the conditions the test needs, and its strength measure, Cramér's V.

Extra: All of this Note is extra material beyond the “which test when” guide; the guide only names the chi-square test.

2. The chi-square statistic

Key point: $\chi^2 = \sum (O - E)^2 / E$ adds up, over all cells, the squared gap between observed and expected counts, each scaled by the expected count.

2.1 Observed and expected counts

Key point: Observed counts come from the data; expected counts are what H_0 predicts for a sample of the same size.

A categorical feature is summarised by counts, one per category (see the [frequency tables Note](#)). For a chi-square test we need two sets of counts:

- **observed counts** O : how many **observations** (records, one row of the data table each) of the sample fall in each category or cell;
- **expected counts** E : how many would fall there, on average, if H_0 were true.

If H_0 is true, O and E differ only by chance. A large mismatch is evidence against H_0 .

2.2 The formula

Key point: Square each gap so that positive and negative gaps both count, divide by E so that a gap of 5 matters more in a small cell than in a big one, and add.

1. **In words:** for each cell, take the gap between observed and expected, square it, divide by expected; add over all cells.
2. **Formula:**

$$\chi^2 = \sum_{\text{cells}} \frac{(O - E)^2}{E}$$

3. **Example:** three cells with $O = 20, 26, 14$ and $E = 15, 33, 12$ (section 4):

$$\chi^2 = \frac{(20 - 15)^2}{15} + \frac{(26 - 33)^2}{33} + \frac{(14 - 12)^2}{12} = 1.667 + 1.485 + 0.333 = 3.48$$

χ^2 is 0 only when every observed count equals its expected count. χ^2 can never be negative.

3. The chi-square distribution

Key point: If H_0 is true, χ^2 follows a chi-square distribution: never negative, skewed to the right, with one parameter, the degrees of freedom.

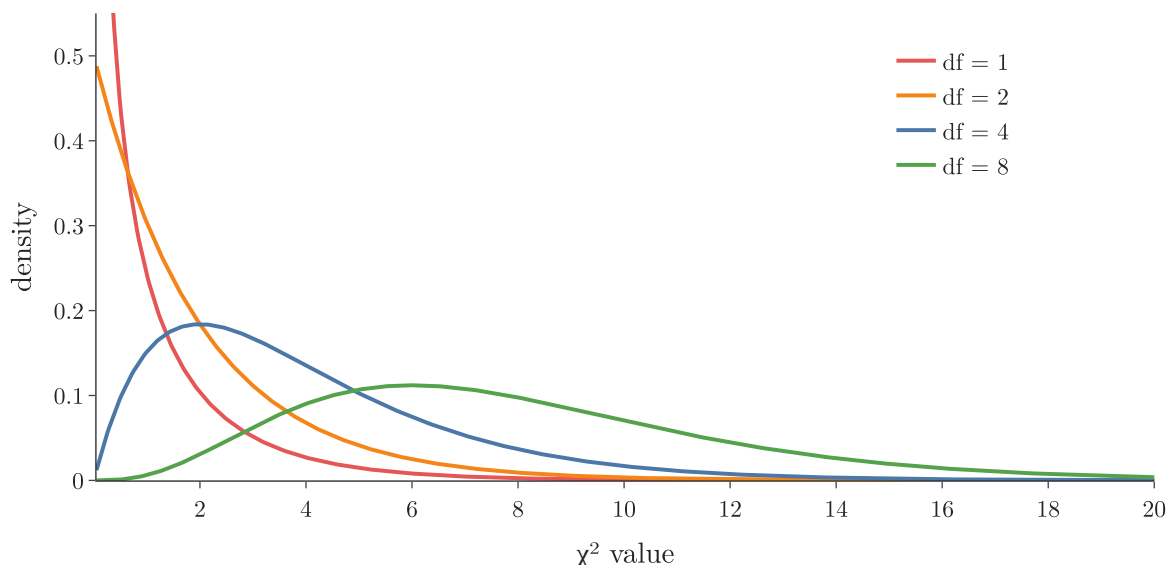


Figure 2: Chi-square densities for four degrees of freedom

The **chi-square distribution** is a continuous distribution on the positive numbers. Like Student's t-distribution (see the [t-procedure Note](#)), it has one parameter, the **degrees of freedom** (df). Figure 2 shows four of them:

- with 1 or 2 df, the density is highest at 0 and falls steadily;
- with more df, the peak moves right and the curve spreads out;
- the mean of a chi-square distribution equals its df (NIST Handbook §1.3.6.6.6).

Only large χ^2 values count as evidence against H_0 , so the p-value is always the **right-tail** area beyond our χ^2 . The 5% critical values are 3.84 for 1 df, 5.99 for 2 df and 7.81 for 3 df.

Extra: Where the distribution comes from: the sum of the squares of k independent standard normal variables follows a chi-square distribution with k df (this is its definition). Each $(O - E)/\sqrt{E}$ is roughly normal for large counts, but the terms are not independent, because the counts must add up to n . Pearson (1900) showed that χ^2 still follows a chi-square distribution for large samples, with one df lost to that constraint. The notebook checks this: 200,000 samples of 60 people drawn with the census shares of section 4 true give a mean χ^2 of 2.00, equal to $df = 2$, and 4.9% of them pass the 5% critical value 5.99.

4. The goodness-of-fit test

Key point: One categorical feature against claimed proportions: expected count = sample size $\times$ claimed proportion, and the degrees of freedom are the number of categories minus 1.

4.1 The question

Key point: Do the age groups in our sample of 60 match a population that is 25% children, 55% adults and 20% elderly?

The **goodness-of-fit test** asks whether the counts of one categorical feature fit a claimed distribution. We use the 60 people of the [choosing a hypothesis test Note](#). Suppose a census says the population is 25% children, 55% adults and 20% elderly.

- H_0 : the age groups occur in the claimed proportions 0.25, 0.55, 0.20;
- H_1 : at least one proportion differs.

4.2 Expected counts and the statistic

Key point: $E = 60 \times 0.25 = 15$ children, $60 \times 0.55 = 33$ adults, $60 \times 0.20 = 12$ elderly; $\chi^2 = 3.48$.

1. **In words:** the expected count of each category is the sample size times its claimed share.
2. **Formula:**

$$E_i = n \times \pi_i, \quad df = k - 1$$

where k is the number of categories.

3. **Example:**

	child	adult	elderly	total
observed O	20	26	14	60
claimed share π	0.25	0.55	0.20	1
expected $E = 60\pi$	15	33	12	60
$(O - E)^2/E$	1.667	1.485	0.333	$\chi^2 = 3.48$

The degrees of freedom are $k - 1 = 2$: once two expected counts are known, the third is fixed by the total of 60.

4.3 The decision

Key point: $\chi^2 = 3.48$ with 2 df gives $p = 0.175 > 0.05$: we fail to reject H_0 .

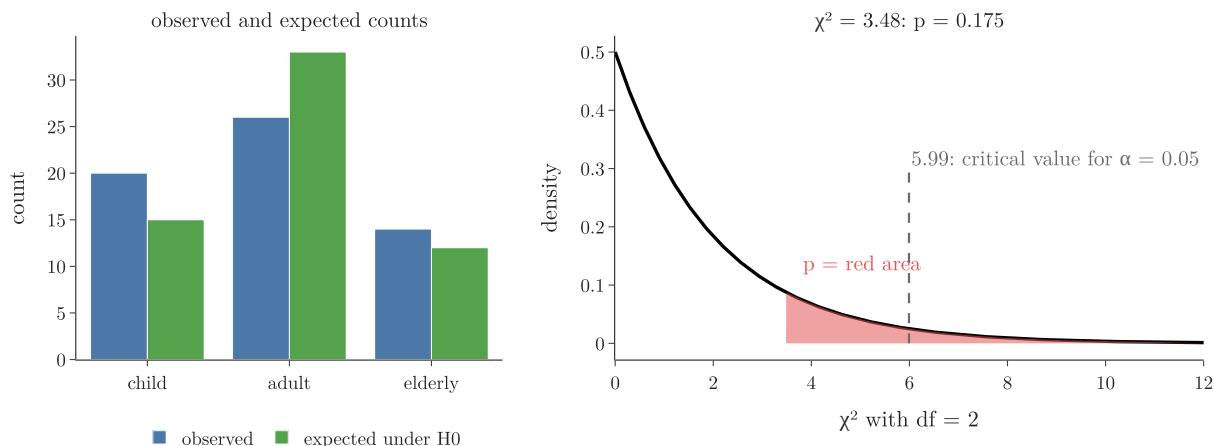


Figure 3: Goodness of fit for the age groups: the counts (left) and the p-value as a tail area (right)

Figure 3 shows both halves of the test. On the left, more children and fewer adults were observed than expected. On the right, $\chi^2 = 3.48$ sits below the critical value 5.99, and the red tail area beyond it is $p = 0.175$.

Since $0.175 > 0.05$, we fail to reject H_0 : the sample is consistent with the census proportions. The gaps on the left are within what chance produces in 60 people.

Python: `chisquare` takes the observed and expected counts.

```
from scipy import stats
```

```
stats.chisquare(f_obs=[20, 26, 14], f_exp=[15, 33, 12])
```

```
# statistic = 3.48, pvalue = 0.175
```

Without `f_exp`, `scipy` assumes equal proportions in all categories.

4.4 Two categories: the proportion test again

Key point: With two categories, the goodness-of-fit χ^2 equals z^2 of the one-sample proportion test, and the p-values are identical.

For gender (26 men, 34 women) against a 50/50 claim, the expected counts are 30 and 30:

$$\chi^2 = \frac{(26 - 30)^2}{30} + \frac{(34 - 30)^2}{30} = 0.533 + 0.533 = 1.067$$

With 1 df this gives $p = 0.302$. The proportion test in the [choosing a hypothesis test Note](#) gave $z = -1.033$, and $(-1.033)^2 = 1.067$, with the same $p = 0.302$. The goodness-of-fit test is the proportion test extended to any number of categories.

Extra: Goodness of fit also checks whether counts follow a named distribution, such as Poisson (see the [Poisson distribution Note](#)). The expected counts then come from the PMF, and every parameter estimated from the data (such as λ from the mean) removes one more degree of freedom: $df = k - 1 - (\text{number of estimated parameters})$. The notebook tests this: 5,000 samples of 200 Poisson counts in 5 categories, with λ estimated each time, give a mean χ^2 of 3.03, matching $df = 3$ rather than 4. Judged with the $df = 4$ critical value, only 2.2% of the samples are rejected instead of 5%; with $df = 3$ the rate is 5.1%.

5. The test of independence

Key point: Two categorical features in a contingency table: if they are independent, each expected count is row total $\times$ column total / grand total, and $df = (r - 1)(c - 1)$.

5.1 The question and the table

Key point: Is the gender mix the same in every age group, or do gender and age group go together?

The **chi-square test of independence** asks whether two categorical features are related. The counts go in a contingency table (see the [contingency tables Note](#)):

	child	adult	elderly	row total
female	12	12	10	34
male	8	14	4	26
column total	20	26	14	60

- H_0 : gender and age group are independent;
- H_1 : they are related.

5.2 Expected counts under independence

Key point: Independence means $P(A \text{ and } B) = P(A)P(B)$; multiplied by n , this gives $E = \text{row total} \times \text{column total} / n$.

Two events are independent when the probability of both is the product of their probabilities (see the [independent events Note](#)). The marginal probabilities come from the totals (see the [joint and marginal probability Note](#)).

1. **In words:** the expected count of a cell is its row total times its column total, divided by the grand total.
2. **Formula:**

$$E = n \times \frac{\text{row total}}{n} \times \frac{\text{column total}}{n} = \frac{\text{row total} \times \text{column total}}{n}$$

3. **Example:** female children: $P(\text{female}) = 34/60$, $P(\text{child}) = 20/60$, so

$$E = \frac{34 \times 20}{60} = \frac{680}{60} = 11.33$$

All six expected counts:

	child	adult	elderly	row total
female	11.33	14.73	7.93	34
male	8.67	11.27	6.07	26
column total	20	26	14	60

The expected table has the same totals as the observed one; only the inside is spread out “as if independent”.

5.3 The statistic and the decision

Key point: $\chi^2 = 2.50$ with 2 df gives $p = 0.29$: no evidence that gender and age group are related.

The contribution $(O - E)^2/E$ of each cell; for female children the contribution is $(12 - 11.33)^2/11.33 = 0.039$:

	child	adult	elderly
female	0.039	0.507	0.538
male	0.051	0.663	0.704

1. **In words:** add the six contributions; the degrees of freedom are $(\text{rows} - 1)$ times $(\text{columns} - 1)$.
2. **Formula:**

$$\chi^2 = \sum \frac{(O - E)^2}{E}, \quad df = (r - 1)(c - 1)$$

3. **Example:**

$$\chi^2 = 0.039 + 0.507 + 0.538 + 0.051 + 0.663 + 0.704 = 2.50, \quad df = (2 - 1)(3 - 1) = 2$$

With 2 df, $p = P(\chi^2 \geq 2.50) = 0.29$. Since $0.29 > 0.05$, we fail to reject H_0 . The elderly group has relatively few men (4 observed, 6.07 expected), but in a sample of 60 a gap like this is common by chance.

Why $(r - 1)(c - 1)$: with the row and column totals fixed, filling $(r - 1)(c - 1)$ cells freely determines all the others. In a 2 by 3 table, choosing 2 cells of the female row fixes the rest.

Python: `chi2_contingency` takes the observed table and returns the statistic, p-value, df and expected counts.

```
import pandas as pd

table = pd.crosstab(people["gender"], people["age_group"])
result = stats.chi2_contingency(table)
result.statistic, result.pvalue, result.dof      # 2.50, 0.29, 2
result.expected_freq                            # the expected table
```

6. Case study: survival on the Titanic

Key point: Sex and class were both strongly related to survival: $\chi^2 = 263$ ($p \approx 10^{-59}$) for sex, $\chi^2 = 103$ ($p \approx 10^{-23}$) for class.

6.1 Sex and survival

Key point: 81 women died where independence predicts 194; 109 men survived where it predicts 222.

The Kaggle Titanic training file has 891 passengers. Survival against sex:

	died	survived
female, observed	81	233
female, expected	193.5	120.5
male, observed	468	109
male, expected	355.5	221.5

The four contributions are 65.4, 105.0, 35.6 and 57.1, so $\chi^2 = 263.1$ with $df = 1$ and $p = 3.7 \times 10^{-59}$. We reject H_0 : survival depended on sex.

Extra: For a 2 by 2 table, `chi2_contingency` applies **Yates' continuity correction** by default: it shrinks each $|O - E|$ by 0.5 before squaring (Yates 1934), which lowers χ^2 and so makes the test slightly more cautious. Here it gives $\chi^2 = 260.7$ instead of 263.1; pass `correction=False` to get the plain statistic. With counts this large the choice does not matter.

6.2 Class and survival

Key point: First class had 136 survivors where independence predicts 83; third class had 119 where it predicts 189.

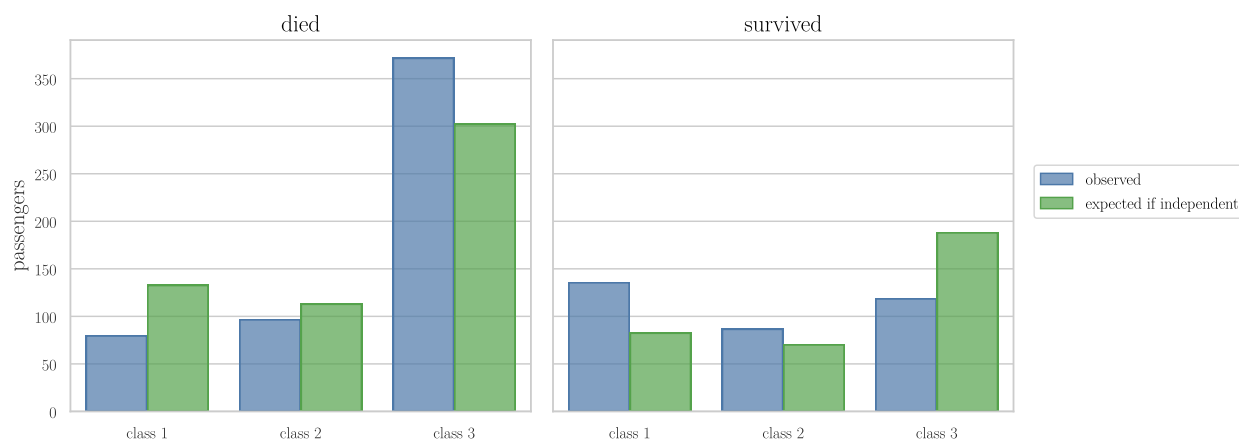


Figure 4: Titanic passengers by class and outcome: observed counts and counts expected under independence

The crosstab of class against survival was drawn as a heatmap in the [bivariate analysis Note](#). Figure 4 compares its observed counts with the expected ones. First class has far more survivors than independence predicts, third class far fewer. The survival rates were 63%, 47% and 24% for the three classes, against 38% overall.

The test gives $\chi^2 = 102.9$ with $df = (3 - 1)(2 - 1) = 2$ and $p = 4.5 \times 10^{-23}$. We reject H_0 : survival depended on class.

7. Conditions and limits

Key point: Use raw counts, independent observations, and expected counts of at least 5 in every cell; a significant χ^2 says that a relationship exists, not how strong it is.

7.1 Conditions

Key point: Counts, independence, and large enough expected counts.

1. **Counts, not percentages.** The formula needs the number of observations in each cell. Percentages or means give a wrong χ^2 .
2. **Independent observations.** Each observation is counted in exactly one cell, and observations do not influence each other. The same person measured twice breaks this.
3. **Large enough expected counts.** The usual rule is that every expected count should be at least 5 (Cochran 1954), because the chi-square distribution is only a large-sample approximation. All six expected counts of section 5.2 are above 5 (the smallest is 6.07). For small 2 by 2 tables, **Fisher's exact test** (`stats.fisher_exact`) avoids the approximation; for bigger tables, merging rare categories helps.

Extra: What goes wrong below the rule. A ruler marked in metres is fine for a room but useless for a grain of rice; the chi-square curve is likewise only a good ruler for large

counts. The notebook draws 200,000 samples with H_0 true, so a correct 5% test should reject 5% of them:

Claimed shares	n	Smallest E	Rejected at 5%
0.25, 0.55, 0.20	60	12	4.9%
0.90, 0.05, 0.05	8	0.4	11.2%
0.90, 0.05, 0.05	10	0.5	2.7%

With large expected counts the test keeps its 5% promise. With two rare categories and tiny samples, the false-alarm rate drifts away from 5%, more than double in one case and about half in the other, so the p-value can no longer be trusted.

7.2 Strength of a relationship: Cramér's V

Key point: Cramér's V rescales χ^2 to a number between 0 (no relationship) and 1 (perfect relationship) (Cramér 1946).

χ^2 grows with the sample size: the same pattern in 10 times as many observations gives a 10 times larger χ^2 . To measure strength, we divide the size out.

1. **In words:** divide χ^2 by the sample size and by one less than the smaller table dimension, then take the square root.
2. **Formula:**

$$V = \sqrt{\frac{\chi^2}{n \times (\min(r, c) - 1)}}$$

3. **Example:** Titanic sex and survival, $\chi^2 = 263.1$, $n = 891$, a 2 by 2 table:

$$V = \sqrt{\frac{263.1}{891 \times 1}} = \sqrt{0.295} = 0.54$$

For class and survival, $V = \sqrt{102.9/891} = 0.34$. Both relationships are significant, but sex was the stronger predictor of survival. In scipy: `stats.contingency.association(table, method="cramer")`.

7.3 Chi-square in machine learning

Key point: Feature selection with `SelectKBest(chi2)` ranks categorical features by how strongly they are related to the target.

Feature selection with `SelectKBest(score_func=chi2)` (see the [pipelines Note](#)) scores each feature against the **target** (the output we predict) with a chi-square statistic: a feature whose counts differ strongly between the classes gets a high score. scikit-learn's version treats each feature's values as counts, which is why it needs values of 0 or more (scikit-learn docs, `chi2`). The score is not the same number as the test of independence. For a one-hot column it

compares only the rows where the column is 1. In the notebook, Titanic sex one-hot encoded gets the scores 170.3 (female) and 92.7 (male); only their sum, 263.1, equals the χ^2 of section 6.1.

8. Summary

	Goodness of fit	Test of independence
Features	one categorical	two categorical
H_0	the categories have the claimed proportions	the two features are independent
Expected count	$n \times \pi_i$	row total $\times$ column total $/n$
Degrees of freedom	$k - 1$	$(r - 1)(c - 1)$
scipy	<code>chisquare</code>	<code>chi2_contingency</code>
Example here	age groups vs census: $\chi^2 = 3.48, p = 0.175$	gender vs age group: $\chi^2 = 2.50, p = 0.29$

- Under H_0 , $\chi^2 = \sum(O - E)^2/E$ follows a chi-square distribution; the p-value is the right-tail area.
- With two categories, goodness of fit is the one-sample proportion test: $\chi^2 = z^2$.
- The test needs raw counts, independent observations and expected counts of at least 5.

9. Sources

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- NIST/SEMATECH (2012). *e-Handbook of Statistical Methods*. itl.nist.gov/div898/handbook. Section 1.3.6.6.6, chi-square distribution.
- Pearson, K. (1900). “On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling”. *Philosophical Magazine* 50(302).
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- Yates, F. (1934). “Contingency Tables Involving Small Numbers and the χ^2 Test”. *Supplement to the Journal of the Royal Statistical Society* 1(2).

10. Key terms

Term	Meaning
Observed count O	The number of observations in a category or table cell
Expected count E	The number of observations a category or cell would hold on average if H_0 were true
Chi-square statistic χ^2	$\sum(O - E)^2/E$: the total mismatch between observed and expected counts
Chi-square distribution	The distribution of χ^2 under H_0 : positive, right-skewed, with one parameter, the degrees of freedom

Term	Meaning
Goodness-of-fit test	A chi-square test of whether one categorical feature follows claimed proportions; $df = k - 1$
Chi-square test of independence	A chi-square test of whether two categorical features are related; $df = (r - 1)(c - 1)$
Yates' continuity correction	A small adjustment to χ^2 for 2 by 2 tables, applied by default in <code>chi2_contingency</code>
Fisher's exact test	An exact test for small 2 by 2 tables, used when expected counts fall below 5